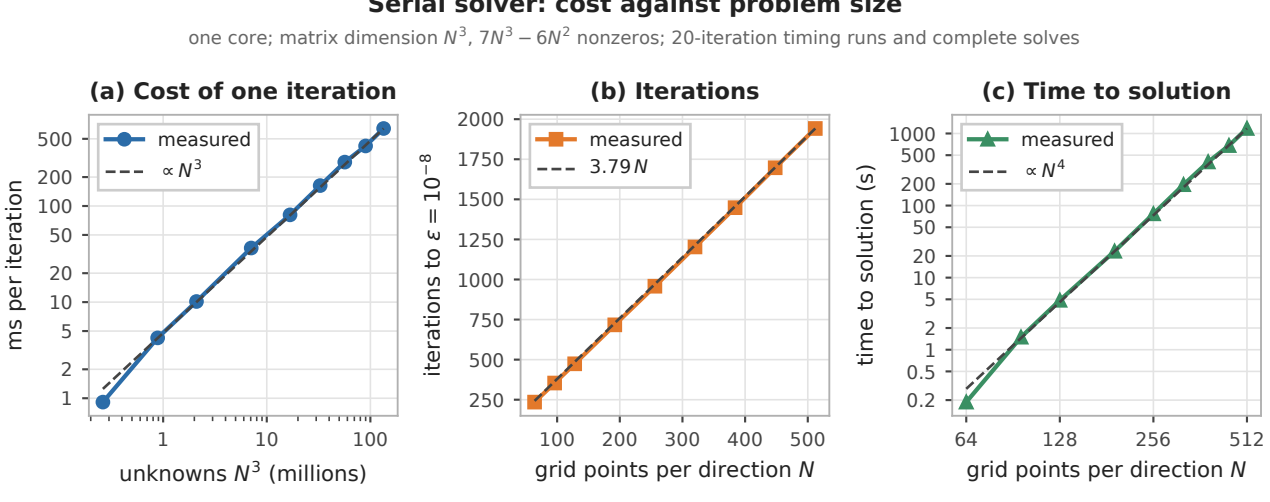
Serial solver: cost against problem size
one core; matrix dimension N^3, $7N^3 - 6N^2$ nonzeros; 20-iteration timing runs and complete solves
(a) Cost of one iteration
ms per iteration
measured
$\propto N^3$
unknowns N^3 (millions)
(b) Iterations
iterations to $\epsilon = 10^{-8}$
measured
$3.79N$
grid points per direction N
(c) Time to solution
time to solution (s)
measured
$\propto N^4$
grid points per direction N